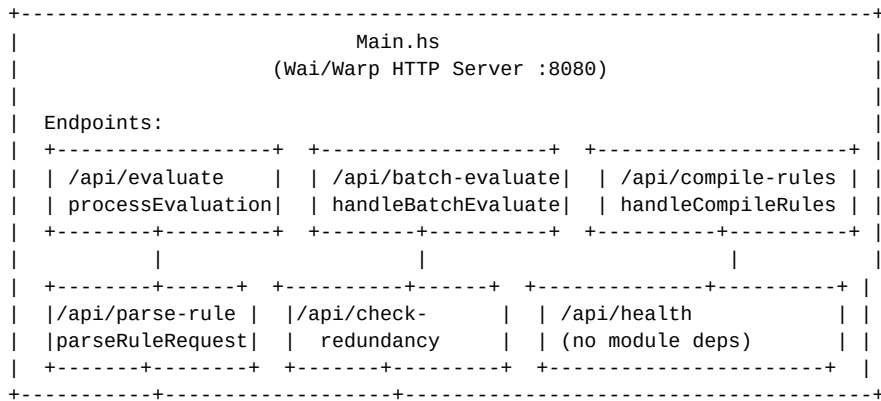
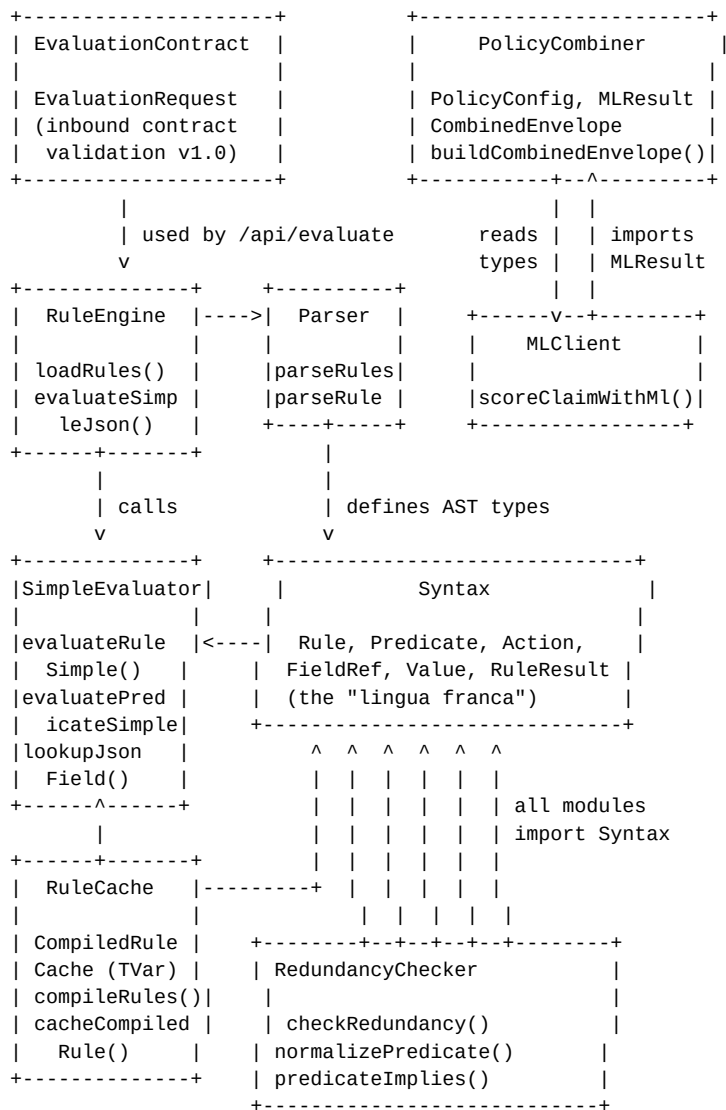


Claims DSL Engine - Module Interaction Flowchart

Main.hs - HTTP Server (Wai/Warp :8080)



Module Dependency Graph



Claims DSL Engine - Module Interaction Flowchart

Request Flow: /api/evaluate (Full Pipeline)

```
HTTP JSON --> EvaluationContract (decode & validate contract v1.0)
--> RuleEngine.loadRules --> Parser.parseRules --> [Rule]
--> RuleEngine.evaluateSimpleJson
    +---> SimpleEvaluator.evaluateRuleSimple (per rule)
        +---> Syntax (AST pattern matching)
--> MLCClient.scoreClaimWithML --> External ML Service
--> PolicyCombiner.buildCombinedEnvelope
    +---> merges DSL RuleResults + MLResult -> CombinedEnvelope
--> JSON response
```

Request Flow: /api/batch-evaluate (DSL-only, Multiple Claims)

```
HTTP JSON --> Main (decode BatchEvaluationRequest)
--> RuleEngine.loadRules --> Parser.parseRules
--> RuleEngine.evaluateSimpleJson (per claim)
    +---> SimpleEvaluator.evaluateRuleSimple
--> JSON response (array of reports)
```

Request Flow: /api/compile-rules (Parse & Cache ASTs)

```
HTTP JSON --> Parser.parseRules --> [Rule]
--> RuleCache.compileRules
    +---> wraps each Rule in CompiledRule (closure over SimpleEvaluator)
--> RuleCache.cacheCompiledRule (STM TVar insert)
--> JSON response (success count)
```

Request Flow: /api/parse-rule (Syntax Check Only)

```
HTTP JSON --> Parser.parseRules --> [Rule] --> JSON AST response
```

Request Flow: /api/check-redundancy

```
HTTP JSON --> Parser.parseRules (candidate + existing)
--> RedundancyChecker.checkRedundancy
    +---> normalizePredicate, predicateImplies
--> JSON response (matches)
```

Module Summary

Module	Role
Syntax	Core AST types -- every other module imports this
Parser	DSL text -> [Rule] AST (Parsec-based)
SimpleEvaluator	Evaluates a Rule against JSON -> RuleResult
RuleEngine	Orchestrator: parse + evaluate all rules + risk level
RuleCache	Thread-safe TVar cache of parsed ASTs
PolicyCombiner	Merges DSL results + ML score -> CombinedEnvelope
MLClient	HTTP client to external ML scoring service
RedundancyChecker	Detects duplicate / subsumed / overlapping rules
EvaluationContract	Inbound request schema with v1.0 contract validation